

**TOPOGRAPHIC LIDAR
SURVEY REPORT**

ISLAND OF NIUE

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1. INTRODUCTION

Woolpert, Inc. (Woolpert) was contracted by the Government of Niue to provide airborne topographic lidar acquisition and processing services for the island of Niue.

Woolpert collected lidar using their Chiroptera 4X (CH4X) topo-bathy lidar sensor, to provide high density topographic and bathymetric lidar. Imagery was also acquired simultaneously with the lidar data using the CH4X system. The system includes a RCD30 80MP 4-band RGBN camera, to provide raw 16-bit radiometrically corrected and normalized imagery.

Details of the survey, data processing, QC and product creation and provided within this report.

1.1. SURVEY AREA

The project area comprised 260km² of topographic data covering the entire island country of Niue. Topographic lidar data were collected to support 100% coverage with at 10 points per square meter in an attempt to penetrate the dense vegetation of Niue.

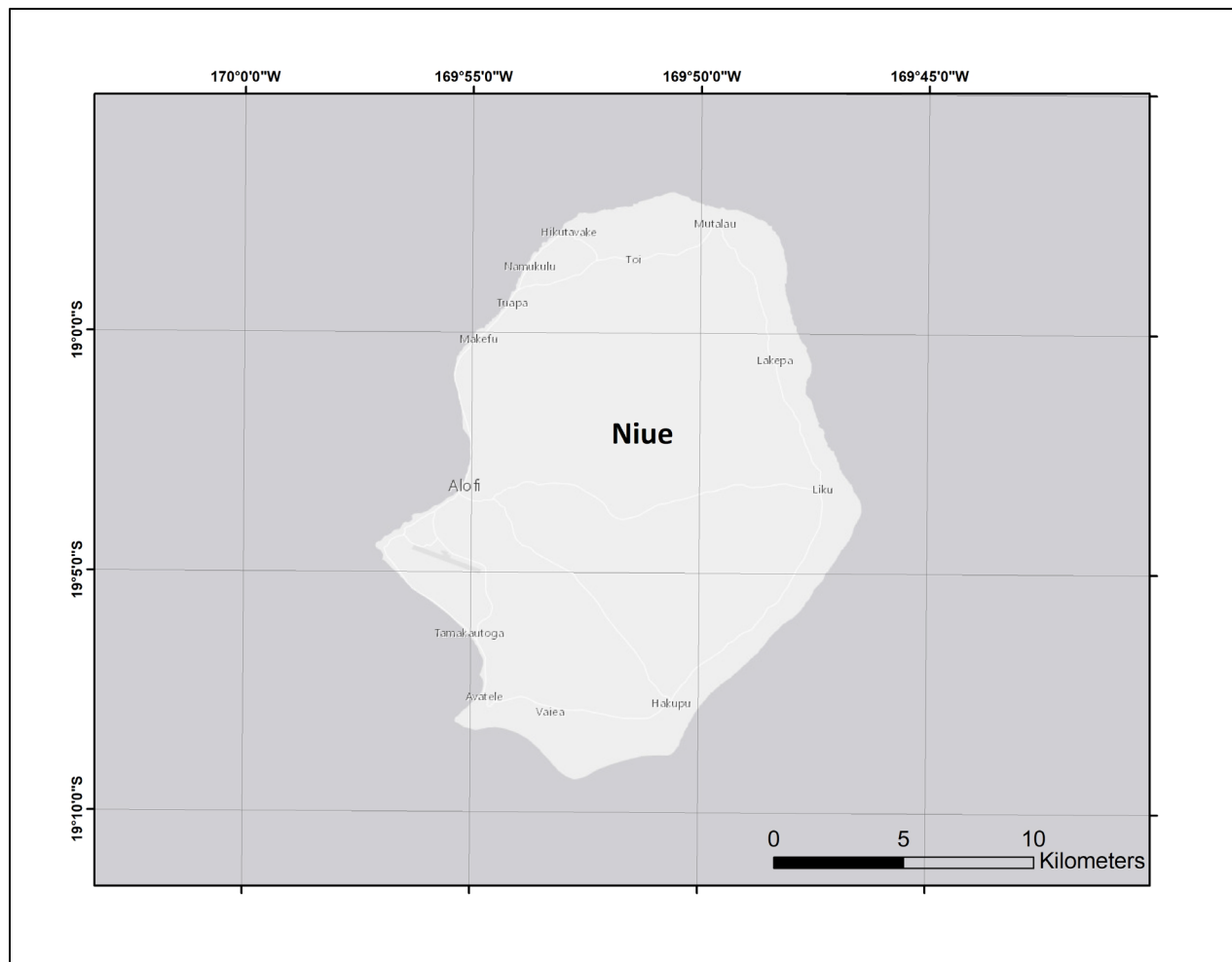


Figure 1: Survey Area

2. DATA ACQUISITION

All lidar and imagery data were acquired using a Chiroptera 4X (CH4X) sensor. The CH4X sensor was mounted in a Leica PAV100 gyro-stabilized mount integrated with a NovAtel SPAN GNSS and LCI-100C IMU. Real time navigation and GNSS/IMU data logging was provided by Leica FlightPro software. Lidar data were logged on the Airborne Hydrography, AB (AHAB) operator console.

2.1. MOBILIZATION

The CH4X was installed in a Reims-Cessna 406 (ZK-XLF) owned and operated by Kiwi Air. A full calibration flight was collected over Niue International Airport (IUE) on 09 January 2020.



Figure 2: Mobilized Survey Aircraft

2.1.1. AIRCRAFT OFFSET SURVEY

Physical mounting offsets between the GNSS antenna, IMU and gyro-stabilized mount were determined through a combination of manual measurements and iterative processing in NovAtel Inertial Explorer software.

Manual measurements were taken from the GNSS antenna to the reference point on the IMU in the CH4X sensor head. These measurements are added to the known offset between the IMU reference point and the rotation center of the gyro-stabilized mount to calculate the preliminary offset between the GNSS antenna and sensor reference point. This preliminary value was then used to seed the post-processing software which, through an iterative computation, uses the dynamic accelerations and rotations during flight to refine the offsets. Once the solution converges, the final offsets are entered into the flight management software and used in subsequent post-processing of the GNSS/IMU data for final trajectories.

Final offsets, shown in the Leica reference frame, are presented in Table 1.

Table 1: Aircraft Offsets

Sensor Head	Lever Arm	X (forward)	Y (right)	Z (down)
CHII (Topo and Shallow Channel)	Reference to GNSS Antenna L1 Phase Center	0.611 m	-0.057 m	-1.220 m
	Reference to IMU	-0.003 m	-0.006 m	-0.296 m
	Reference to IMU Rotation	0°	0°	90°

2.2. LIDAR CALIBRATION

Field calibration of the CH4X system is carried out to eliminate systematic errors by calculating corrections for boresight errors, scanner angle errors, remaining IMU angle errors and any necessary internal timing errors. In order to verify or compute the field calibration, the following lines are flown (Figure 3):

- 2 x Line A over mixed terrain with flat or gentle slopes and features such as peaked roof buildings (1 x each direction)
- 1 x Line B offset +50% from Line A in one direction
- 1 x Line C offset -50% from Line A in the same direction as Line B
- 2 x Line D orthogonal to previous lines (1 x each direction)

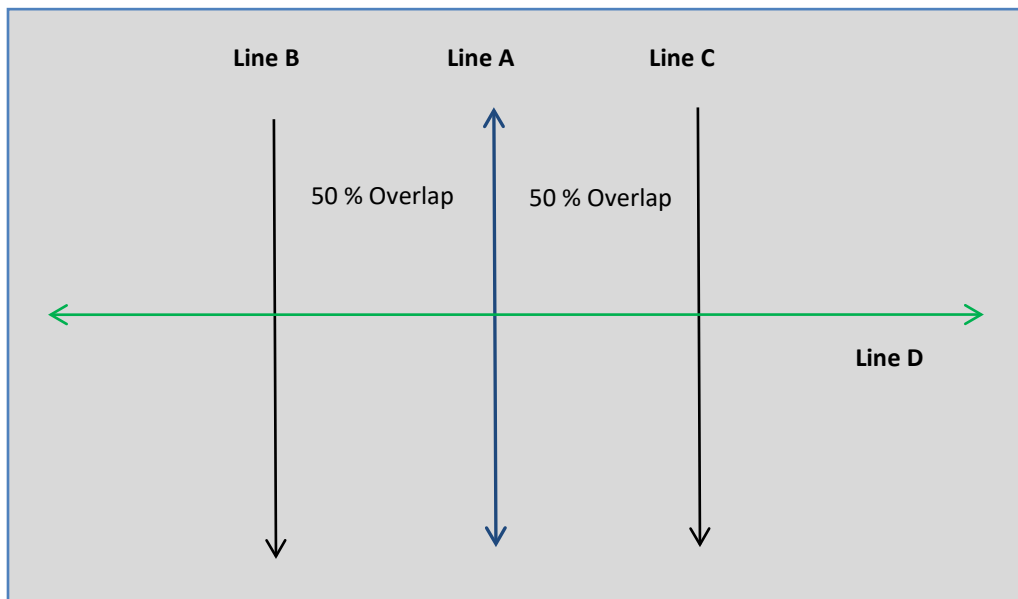


Figure 3: Schematic of HE3 Calibration Lines

A set of calibration lines were acquired at 500m and 400m altitude over the Niue International Airport (IUE). Persistent high clouds in the area would not allow for the collection of lines at higher altitudes.

Calibration values are calculated using the automatic calibration routine within the Leica Lidar Survey Studio (LSS) software. This utility first identifies patches or areas of gentle slope within the overlap region of all the lines to use for calibration. Patch selection prevents areas of vegetation, side of cars or buildings, from being used in the calibration process. Next, the utility compares the front side and back side of the elliptical scan within the same line, as well as comparing all lines to each other, to identify suitable calibration parameters such that data within the patches match. The procedure is iterative and continues until the best possible solution is computed.

Calibration for each channel (topo and shallow) is done independently. For each channel, calibration was computed using 500m altitude lines. The 400m lines were then used for verification.

At each step of the calibration process, quality assurance is conducted to ensure values being calculated are valid. This is done using the Leica LSS Quality Control Utility. Two types of checks are done; firstly, the front scan is compared to the back scan for every line. Secondly, a single line is chosen as a baseline and is compared to every other line. We would expect the average errors from both of these checks to be small; less than 2cm (Table 4 and Table 5). In addition, the data is visually reviewed. In particular, features are studied to ensure lines from different directions show structures in the same position, in other words, verifying horizontal accuracy is maintained. These tests all provide assurance of relative accuracy. Parameters for the calibration flight are listed in Table 2 below.

Table 2: Mission Parameters for Calibration Flight

Flight Speed (knots)	Altitude (m)	Scan Rate (kHz)	Scan Line Speed (rpm)	Overlap (%)
130	400m, 500m	300 kHz at 400m, 500 kHz at 500m (topo), 140 kHz (shallow)	4220 (topo), 3100 (shallow) at 400m 4200 (topo), 3000 (shallow) at 500m	50

Ground truth is not used within the automatic calibration routine; however, ground truth can be used to verify absolute accuracy.

For this project, calibration lines were acquired at the Niue Airport (IUE). Woolpert acquired GNSS RTK ground truth data over the parking areas of airport using Trimble R10 GNSS receivers. To provide accurate coordinates for comparison, a temporary control point (RM1) was established on the airport grounds which served as the base for RTK observations. GNSS observations were collected over multiple sessions on multiple days and the data uploaded to Trimble's RTX service, to compute an accurate coordinate for the control point. The average of 4 observation sessions was used as the final coordinate (Table 3).

Table 3: Control Point Coordinate (RM1)

Control Point	Latitude	Longitude	Height (m)	Datum	RMS (m)
RM1	19° 04' 35.10241" S	169° 55' 37.27601" W	88.102	ITRF14	0.013

Results from the calibration are provided in Table 4 below. Results are good and indicate that calibration was successful. Values computed will be used for the entire project.

Table 4: Calibration QA Results

Test		Topo 500m	Topo 400m	Shallow 500m	Shallow 400m
Front to Back Scan Comparison	Average Error (m)	-0.0004	-0.0032	-0.0025	-0.0200
	Std. Dev. of Error	0.0004	0.0008	0.0014	0.0014
Line to Line Comparison	Average Error (m)	0.0099	-0.0063	0.0064	-0.0030
	Std. Dev. of Error	0.0015	0.0032	0.0032	0.0083

A comparison to the ground truth was also conducted. Results presented in Table 5 below show data is well within expected accuracy specifications.

Table 5: Calibration Ground Truth Comparisons

	Topo		Shallow	
	500m	400m	500m	400m
Average dz (m)	-0.0073	0.0057	-0.0055	-0.0118
Root Mean Square (m)	0.0142	0.0187	0.0113	0.0385

As discussed, resulting data is also analyzed visually. Results can be seen in the figures below.

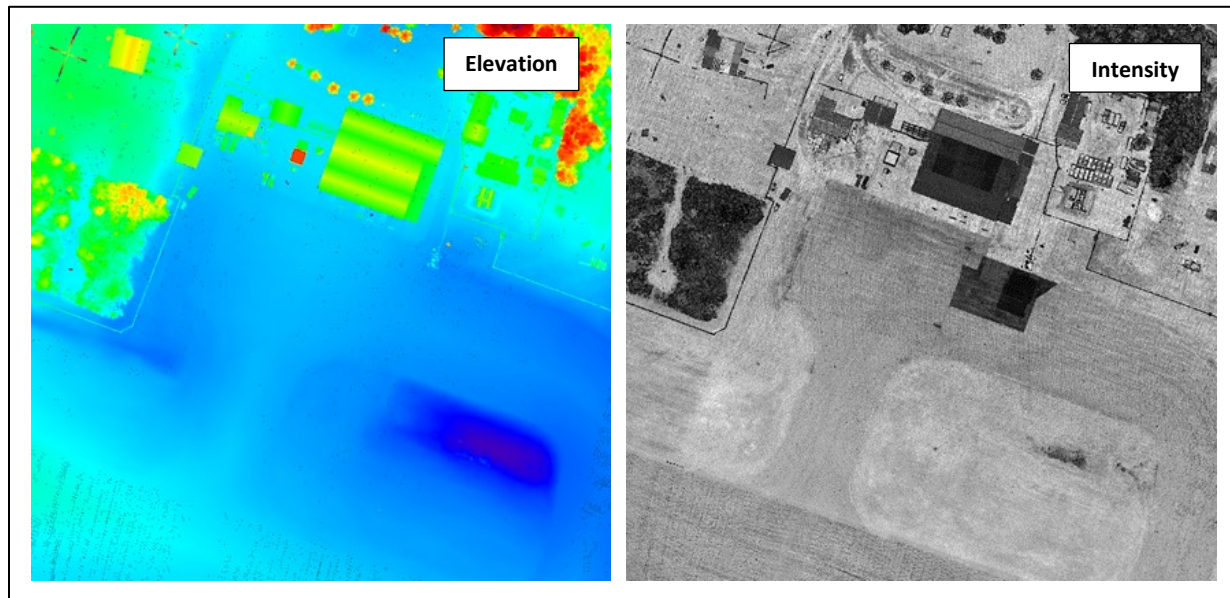


Figure 4: Calibration Site Niue International Airport

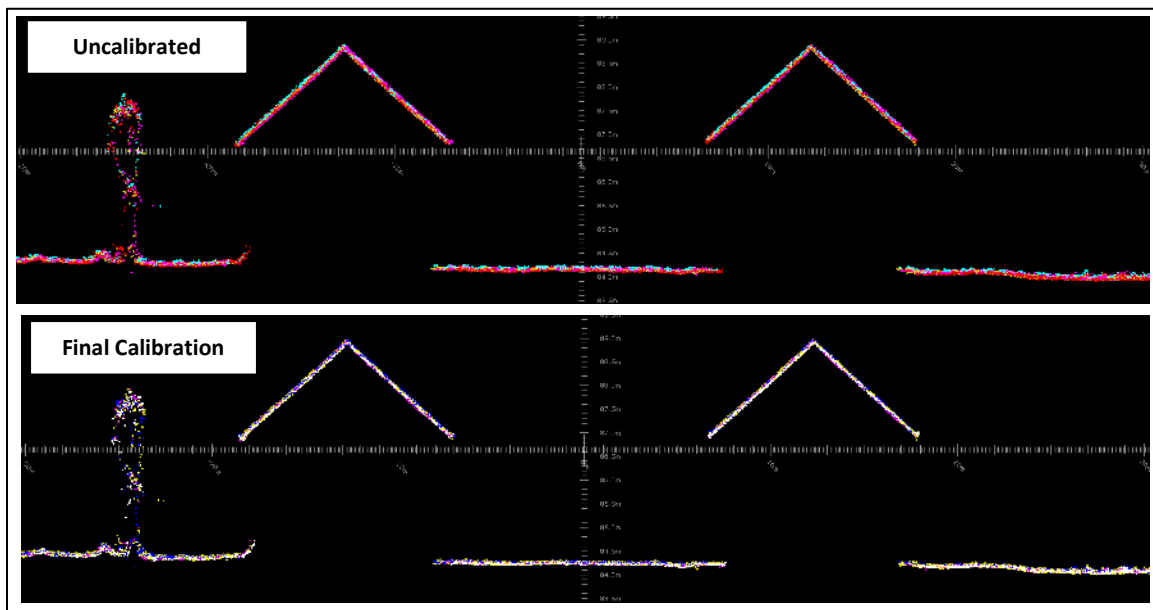


Figure 5: Profile View Colored by Flight Line - Topo Channel Calibration

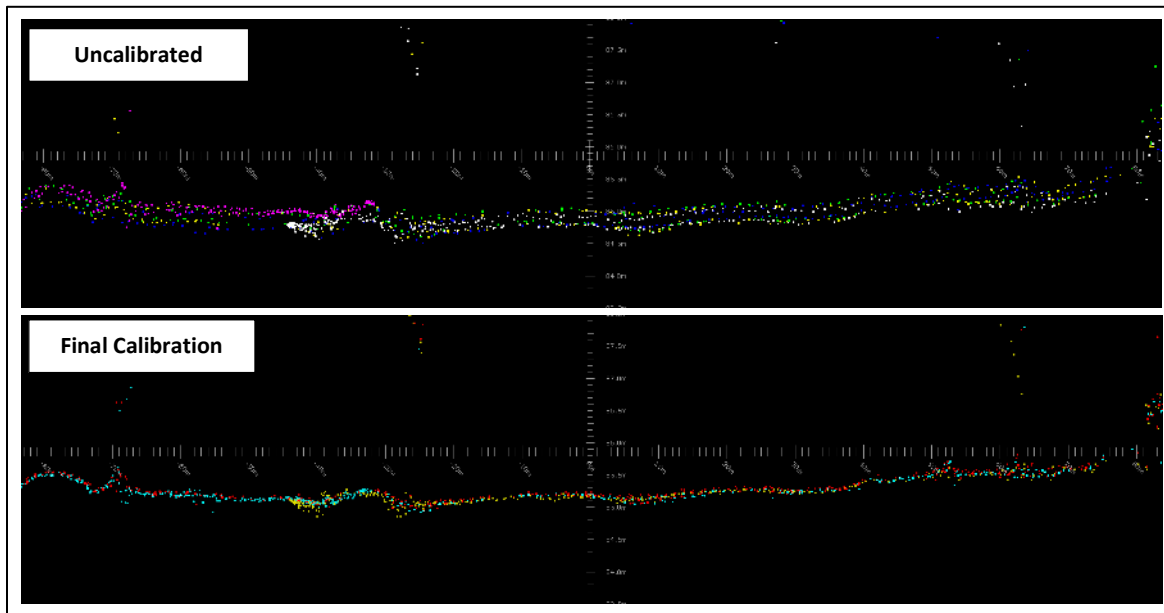


Figure 6: Profile View Colored by Flight Line - Shallow Channel Calibration

2.3. IMAGE CALIBRATION

Field calibration of the RCD30 camera is carried out to eliminate systematic errors by calculating corrections for the principal point offset and misalignment angles (Omega, Phi Kappa). Misalignment angles are essentially the mechanical misalignment of the IMU and the camera sensor axes.

Leica's IPAS CO+ was used to finalize the camera calibration. It uses orthogonal lines flown in both directions over an area containing buildings and features, such as painted lines. In this case, orthogonal lines A and D (Figure 3) from the calibration flight were used.

IPAS CO+ has an automated point matching (APM) feature that identifies the same point in overlapping images and automatically iterates to compute final misalignment and principal point offset (PPO) parameters, which are provided in the table below.

Table 6: RCD30 Camera Misalignment and PPO Parameters

Parameter	X	Y	Z
Lever Arms (m)	0.000	-0.115	0.166
Rotation (deg)	0 °	0 °	90 °
Misalignment (deg)	-0.08146	-0.02264	0.11985
PPO (mm)	0.0412	-0.0109	N/A

The imagery was then reprocessed in the HxMAP software with the new calibration parameter file, and fixed features on the ground visible in the imagery such as the end of paint marks or the intersection of concrete pads in the calibration area were analyzed to ensure imagery from each line matched. Results from the analysis of 33 tie point locations, using imagery acquired at 500m, provides the RMS values given in Table 7. RMS values are low and indicate the accuracy of the imagery is within expected results.

Table 7: RMS Values for Calibrated RCD30 Imagery

No. of Comparison Locations	33
Altitude of Imagery (m)	500
RMS XY (m)	0.0487



Figure 7: Horizontal Ground Truth Locations used in RCD30 Accuracy Assessment

2.4. SURVEY OPERATIONS

A summary of the daily operations is shown in Table 8.

Table 8: Summary of Daily Operations

Flight (UTC Dates)	Engine Time	Air Time	Flown		Reflown		Comments
			km	%	km	%	
2020-01-07A	1:41:00	1:11:00	110.7	9.3%			Niue
2020-01-08A	4:56:00	4:40:00	680.5	57.1%			Niue
2020-01-08B	3:09:00	2:55:00	400.9	33.6%	27.8	2.3%	Niue
2020-01-09A	1:21:00	1:01:00					Niue Calibration
TOTAL	11:07:00	9:47:00	1192	100%	28	2.3%	

2.4.1. THE CHIROPTERA 4X SENSOR

All lidar and imagery data were acquired using a CH4X sensor (Figure 8). The CH4X is a 1.5x secchi depth system and is the latest generation topographic and bathymetric lidar sensor commercially available. The system provides denser data than previous traditional bathymetric lidar systems. It is unique in its ability to acquire bathymetric lidar, topographic lidar and 4-band multi spectral digital camera imagery simultaneously. Therefore, any land data will always be illuminated by the topographic laser, and any water data will be illuminated with the bathymetric laser during any single flight line.

The CH4X is a two channel lidar system with independent lasers that collect topographic and shallow bathymetric data. The CH4X provided 500 kHz topographic data and an effective 140 kHz shallow bathymetric data. 4-band 80 MP digital camera imagery was also collected simultaneously with the sensor's RCD-30 camera.

The bathymetric and topographic lasers are independent and do not share an optical chain or receivers, so they are optimized for their specific function. As with any bathymetric lidar, maximum depth penetration is a function of water clarity and seabed reflectivity. The CH4X is designed to penetrate to approximately 1.5 times the secchi depth. This is also represented as $D_{max} = 2.44/K$, where K is the diffuse attenuation coefficient, and assuming K is between 0.1 and 0.3, a normal sea state and 15% seabed reflectance.

Both the topographic and bathymetric sub-systems use a palmer scanner to produce an elliptical scan pattern of laser points with a degree of incidence ranging from +/-14° (front and back) to +/-20° (sides), providing a 40° field of view. This has the benefit of providing multiple look angles on a single pass and helps to eliminate shadowing effects. This can be of particular use in urban areas, where all sides of a building are illuminated, or for bathymetric features such as the sides of narrow water channels or features on the seafloor such as smaller objects and wrecks. It also assists with penetration in the surf zone where the back scan passes the same ground location a couple of seconds after the front scan, allowing the areas of whitewater to shift.

The bathymetric lasers are diode pumped class 4 lasers which operate in the green spectrum. Full waveform data is acquired for every pulse. The topographic laser operates in the near infra-red spectrum at 1064nm. Up to 4 returns per pulse are acquired from each laser.



Figure 8: Chiroptera 4X Bathymetric Lidar Sensor Installation

For this project, the flight parameters shown in Table 9 were used. Flight lines were planned to provide 100% coverage with a 15% sidelap. A second set of lines were then planned, offset 50% from the first set, to provide overall 200% coverage over all survey areas. Flight parameters used easily exceed the requirements for the survey to produce 2m grids and meet LINZ Specifications.

During acquisition, flight lines are shown on a pilot's display, and the aircraft is controlled by the pilots at all times. The CH4X system includes a NovAtel SPAN GNSS system with an LCI-100C IMU for aircraft position and orientation. The IMU is in the main Chiroptera sensor head, which includes the topo channel, shallow channel and RCD30 camera. Information from this IMU is also used in real-time by the PAV100 gyro-stabilized mount to compensate for deviations in pitch and roll. Aircraft bank angles were restricted to 20° to avoid any potential GNSS dropouts. No flights were planned if the PDOP was expected to go above 3.0.

Data were monitored for quality during acquisition using the Operators Console running on the AHAB collection computer. The operator monitored system status of the scanners and receivers, waveforms, camera images, data coverage, flight lines and the health of the navigation system.

All data were recorded to a removable solid-state hard disk. At the end of the flight the hard disk was removed and taken to the field office where data was copied on to backup disks for transmittal back to the main processing office. Data was reviewed daily in the field for quality and coverage.

Table 9: CH4X Survey Flight Parameters

Aerial System	Leica Chiroptera 4X (topo-bathy)
Nominal Survey Altitude	500m
Nominal Survey Speed	130 knots
System Planned Sidelap	15%
Lidar	
Scan Angle	$\leq 40^{\circ}$ (+/-20° from Nadir)
Nominal Swath Width	350m
PRF	500 kHz
Points Per Square Meter (Topo)	≥ 10
Returns Collected Per Laser Pulse	Up to 4
Intensity Range	0 – 65535 (16-bit)

2.4.2. POSITIONING

Position and orientation data were acquired in the aircraft using a NovAtel SPAN with LCI-100C IMU. All data were post-processed using NovAtel Inertial Explorer software to provide a tightly-coupled position and orientation solution.

A single GNSS base station was used to control trajectory processing providing all final trajectories on ITRF2014. Table 5 contains the location of the base station used for the project.

3. DATA PROCESSING

An overview of Woolperts established CH4X processing workflow is presented in Figure 8. Initial data coverage analysis and quality checks to ensure there were no potential system issues were carried out in the field prior to demobilization of the sensor. Final processing was conducted in Woolperts offices.

In general data were initially processed in Leica's Lidar Survey Studio (LSS) using final processed trajectory information. LAS files from LSS were then imported to a Terrascan project where spatial algorithms were used to remove noise and classify bare earth/ground. Manual review was conducted in both Terrascan and LP360 prior to generating final products.

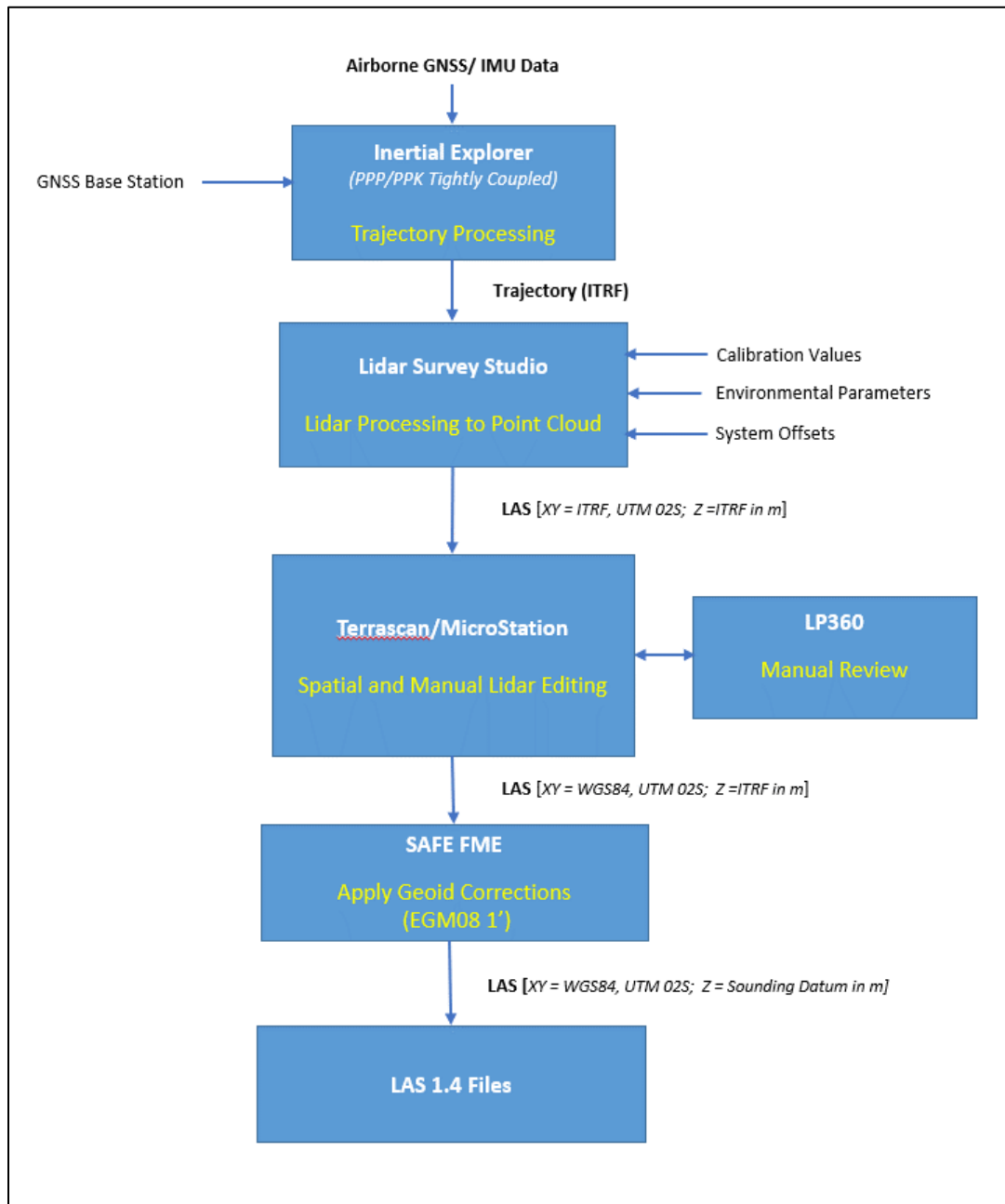


Figure 9: Overview of Processing Work Flow

3.1. POSITION

Final trajectory data were post processed in NovAtel Inertial Explorer. Lever arms, shown in the NovAtel reference frame, are presented in Table 10. Inertial Explorer accounts for the fixed offset between the reference point and IMU and uses a multi-pass algorithm to compute a tightly-coupled solution. A single GNSS base station was used for processing, as described in Section 2.2. Trajectory processing logs are provided in Appendix B.

Table 10: Inertial Explorer Offsets

Sensor Head	Lever Arm	X (right)	Y (forward)	Z (up)
Chiroptera (Topo, Shallow, Camera)	Reference to GNSS Antenna L1 Phase Center	-0.052 m	0.614 m	0.924 m
	Reference to IMU Rotation	0 °	180 °	0 °

3.2. IMAGERY

Imagery data collected with the RCD30 camera were extracted from the raw compressed airborne format to 8-bit 4-band (RGBN) images using Leica's HxMAP software.

Individual images were rectified by HxMAP using a 1m resolution Digital Surface Model (DSM) created from the preliminary topo lidar data. Rectification was accomplished using direct georeferencing (making use of the final post-processed aircraft trajectory, along with the camera interior and exterior camera information) without use of any Aerial Triangulation and ground control. This was necessary due to the remoteness of the area, but also easily met the required accuracy specifications for the imagery.

All individual rectified images for an area were used in Trimble's OrthoVista software to create the final 10cm resolution tiled image mosaics in geotiff format. The tiles are 500m x 500m in size. Geotiffs were also converted to 3 band ECW files.

Image tiles are named based on unique identifiers in the Easting and Northing labels for an area. For example, image 61308815 has a bottom left corner of **613000mE** and northing **7 881 500mN**.

3.3. LIDAR

3.3.1. RAW DATA PROCESSING

Lidar processing was conducted using the Leica Lidar Survey Studio (LSS) software. Calibration information, along with processed trajectory information were combined with the raw laser data to create an accurately georeferenced lidar point cloud for the entire survey in LAS v1.4 format. All points from the topographic and bathymetric laser include 16-bit intensity values.

During this LSS processing stage, an automatic land/water discrimination is made for the bathymetric waveforms. This allows the bathymetric (green) pulses over water to be automatically refracted for the pulse hitting the water surface and travelling through the water column, producing the correct depth. Another advantage of the automatic land/water discrimination is that it permits calculation of an accurate water surface over smaller areas, allowing simple bathymetric processing of smaller, narrower streams and drainage channels. Sloping water surfaces are also handled correctly.

Prior to processing, the hydrographer can adjust waveform sensitivity settings dependent on the environment encountered and enter a value for the refraction index to be used for bathymetry. The index of refraction is an indication of the water type. Values used for sensitivity settings and the index of refraction are included in the LSS processing settings files. A value of 1.3423 was used for the index of refraction, indicating salty water.

In the field, default waveform sensitivity settings were used for processing. In order to determine the optimal waveform sensitivity settings for final processing, sample areas were selected and processed with multiple different settings, to iteratively converge on the best possible settings. This is done by reviewing the processed point cloud and waveforms within sample areas. A sample waveform is provided in Figure 10, while a sample LSS editing screen is provided in Figure 11. Settings affect which waveform peaks are classified as valid seabed, and which peaks are classified as noise. Optimal settings strike a balance between the amount of valid data that is classified as seabed bottom, and the amount of noise that is incorrectly classified due to peaks in the waveforms. Ideally all valid data is selected, while only a small amount of noise remains to be edited out. Once optimal threshold settings were chosen, these were used for the entire project.

It is important to note that all digitized waveform peaks are available to be reviewed by the hydrographer; both valid seabed bottom and peaks classed as noise. This allows the hydrographer to review data during Terrascan and LP360 editing for valid data such as objects that may have been misclassified as noise.

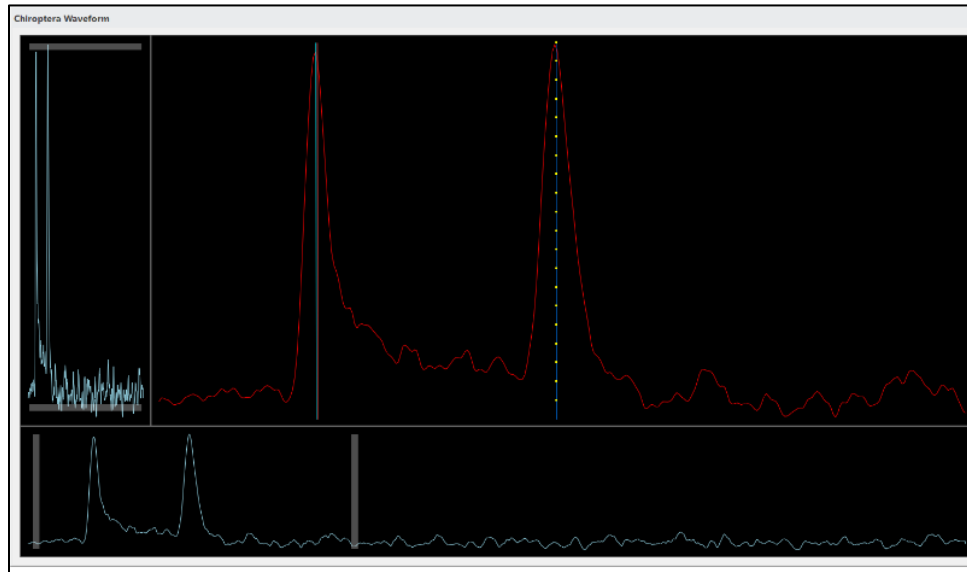


Figure 10: Sample Waveform in Shallow Water

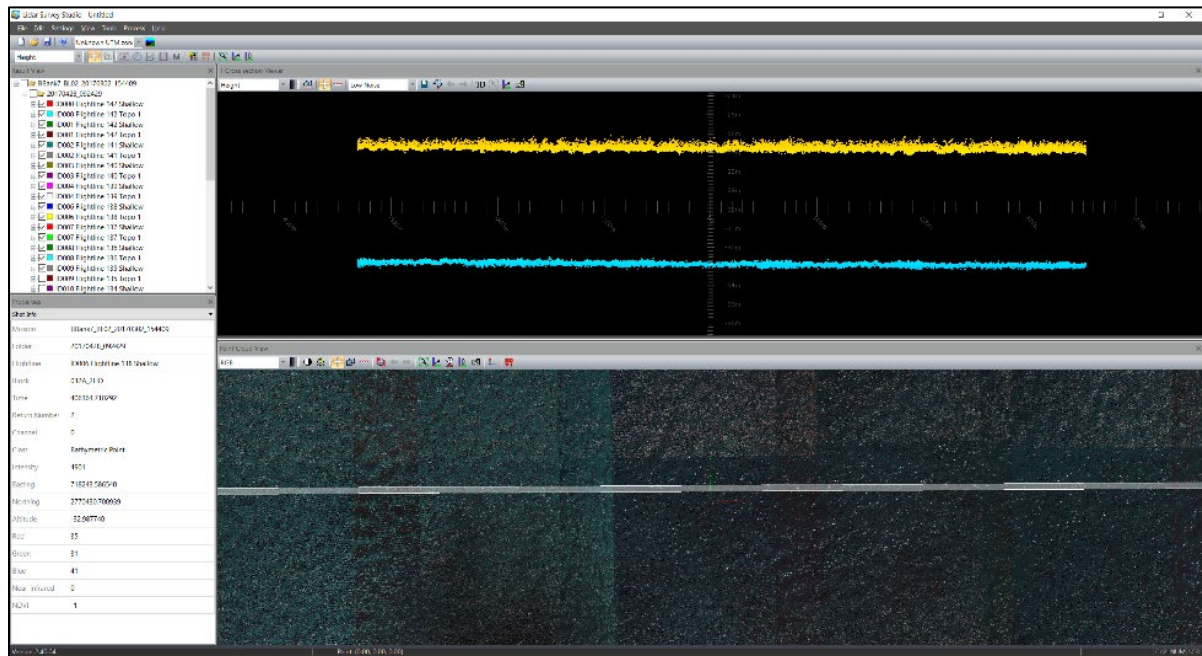


Figure 11: Sample LSS Processing Screen

LSS processing produced LAS files in 1.4 format.

Additional QC steps were performed in LSS prior to import to Terrascan. Firstly, the derived water surface was reviewed to ensure a water surface was correctly calculated for all bathymetry channels. No significant issues were apparent.

Spot checks were also made on the data to ensure the front and back of the scans remained in alignment and no calibration or system issues were apparent prior to further data editing in Terrascan.

LSS Processing Logs are provided in Appendix B. The logs indicate the calibration files used and processing session that data were output to.

LSS stores data in multiple LAS files for a single flight line. Each file corresponds to a single .dat file from the raw airborne data. Woolpert merged these multiple files into a single file per flight line and moved data into a standard class definition in preparation for data editing using Woolpert proprietary scripts within SAFE's FME software.

3.3.2. LIDAR DATA EDITING

After data were processed in LSS and the data integrity reviewed, data were organized into tiles within a TerraScan project. Data classification and spatial algorithms were applied in Terrasolid's TerraScan software. Customized spatial algorithms, such as isolated points and low point filters, were run to remove gross fliers in the bathymetric and topographic data. A grounding algorithm was also run on the topographic data to distinguish between points representing the bare earth, and other valid topo lidar points representing features such as vegetation, buildings, and so forth.

Topographic data were reviewed manually to reclassify any valid ground incorrectly identified by the automated routines in LSS as invalid, and vice versa. In addition, any topo points over the water were reclassified to Water to correct the ground representation. Bathymetric data were reviewed to remove obvious misclassifications in order to provide a general representation of the seabed. As this is a topographic survey, bathymetric data were not reviewed to hydrographic charting standards. Manual editing was conducted both in Terrascan and LP360. Steps for manual editing included:

- Review topo ground points in areas of gaps or spikes.
 - Add points to ground (Class 2) from the bathy or topo laser if points are available to fill gaps in the ground model, typically in thick vegetation.
 - Re-class any noise in the ground class to Topo Unclassified (Class 1) if valid vegetation or other feature, or Noise if the point is not valid (Low Noise (Class 7) or High Noise (Class 18)).

With the new Chiroptera 4X having denser data from the bathymetric laser, an unexpected benefit was improved penetration with the bathy laser in dense vegetation. This likely occurs because the laser is higher powered with a wider field of view. Data from the bathy laser was used to fill the ground on land, where no topo ground points existed. These points remain in Class 2 in the final LAS files, but are coded with the Scanner 2 (bathy) scanner number. An example is provided from an area of dense vegetation (Figure 12), where the upper cross section shows only topo laser ground points in grey and the lower cross section shows both topo points in grey while bathy laser ground points are green, providing excellent ground coverage.

Once editing was completed in TerraScan, 2m resolution Digital Elevation Models (DEMs) were created using LASTools. These were reviewed in QT Modeler for any remaining issues.

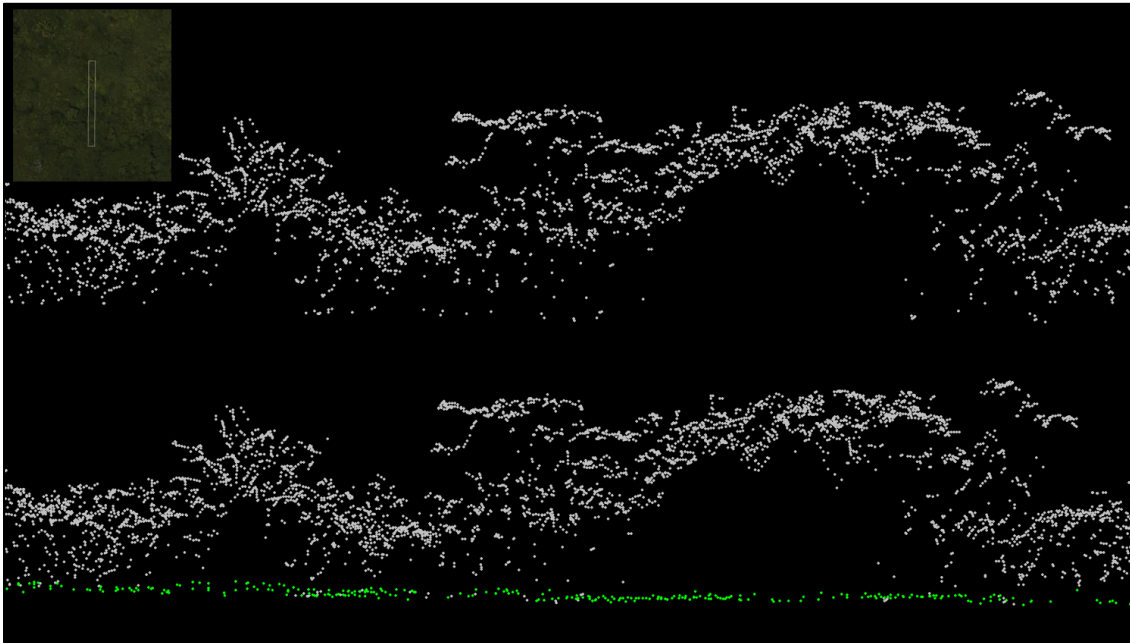


Figure 12: Bathymetric Laser Data (Green) Penetrating to the Ground in Dense Vegetation

3.3.3. LIDAR DATUM CONVERSIONS

All lidar editing in Terrascan and LP360 was conducted with the lidar data elevations on the ellipsoid. This allowed QC checks to be conducted against control which also existed on the ellipsoid. In addition, one set of LAS files was required to be delivered on the ellipsoid. Once the major editing steps were concluded, two sets of LAS tiles were prepared for delivery:

- One set on the ellipsoid
- One set with a static shift applied (described below) as part of the requirements to move the ellipsoid elevations to Sounding Datum

Since lidar is acquired on the ellipsoid, time-based water level information cannot be directly applied to the lidar information. Therefore, the data must be transformed from the ellipsoid to an orthometric datum using a suitable geoid model. Then a transformation from the orthometric datum to the required tidal datum must be applied. Since the geoid model will roughly follow global mean sea level, in areas without complex tidal regimes a simple shift is applicable to move between the orthometric datum and the tidal datum. In more complex tidal regimes, a model must be developed to move between the orthometric datum and any given tidal datum. An example of a simple conversion, based on the numbers for Niue, is provided in Figure 13.

In the case of Niue, the global EGM08 1' geoid model was used and revealed a stable tidal regime, allowing the application of a static shift from the orthometric datum to the tidal datum for each of the individual areas.

Based on the tidal datum analysis provided by LINZ for the 2018 survey, the offsets provided in Table 11 were applied to the lidar data for the project (where N represents the EGM08 1' geoid model offset and therefore varies by location).

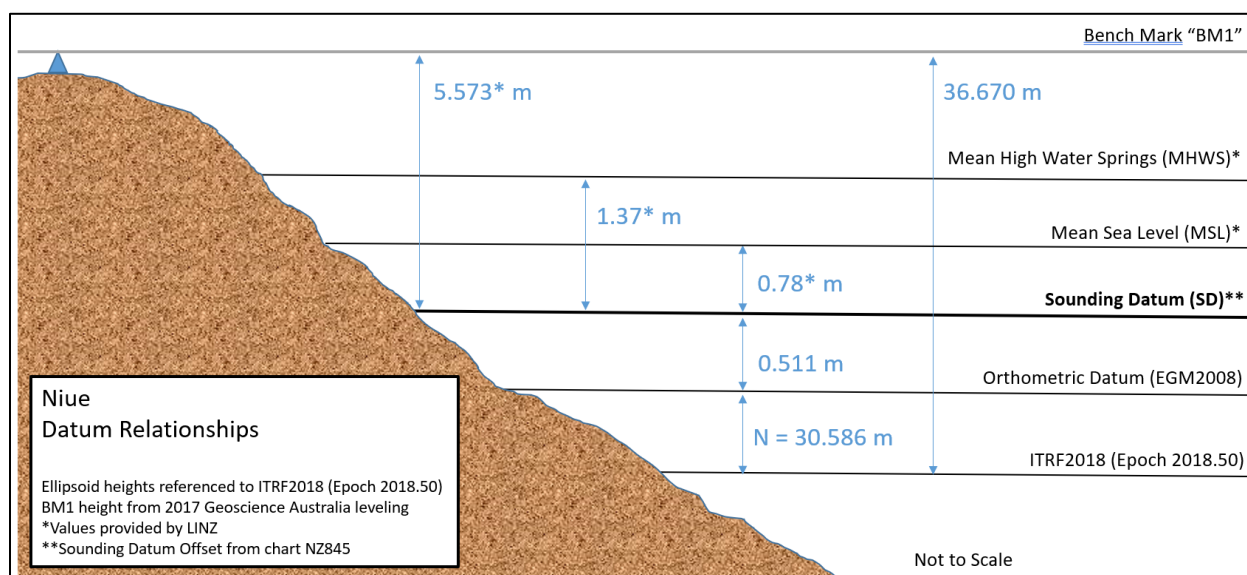


Figure 13: Datum Relationships at Niue Benchmark

Table 11: Shift from Orthometric Datum to Sounding Datum

Geoid Model Applied	Static Shift from Ortho to Sounding Datum (m)
EGM08 1'	-0.511

The sounding datum LAS files were exported from Terrascan with the static shift applied. These files were then run through SAFE FME software to apply the EGM08 1' geoid model to the vertical elevations. No horizontal transformation was applied.

4. QUALITY CONTROL

Quality control is carried out through every phase of the project. Several checks were used to ensure data integrity and quality was maintained.

4.1. CALIBRATION

This is fundamental to good data accuracy. Calibration is discussed in detail in Section 2.2.

4.2. ONLINE CHECKS

The airborne operator monitored system status of the scanners and receivers, waveforms, camera images, data coverage, flight lines and health of the navigation system during data acquisition. Flight logs are maintained during data acquisition. Logs not only track lines acquired, but also any relevant information on weather or water clarity, instances when sensor issues occur and so on. These logs are a valuable resource during processing. They are provided in Appendix A.

4.3. POSITIONING

During acquisition, aircraft bank angles were restricted to 20° to avoid any potential GNSS dropouts. No flights were planned if the PDOP was expected to go above 3.0.

4.4. COMPARISON TO ADJACENT LINES

Throughout data editing adjacent survey lines of data are compared to ensure there are no data busts, or system artifacts. During processing Terrasolid's TMatch software is run to examine the Delta Z differences between overlapping lines. If differences are greater than 0.03m, then a simple Z correction is applied per flight line to remove any vertical differences between flight lines. TMatch can then be run again once all corrections are applied to ensure adjacent lines agree within specification.

Average line to line Z mismatch is provided in Table 12. Differences show good agreement between lines.

Table 12: Z Differences Between Adjacent Lines

Survey Area	Average Z Mismatch (m)
Niue	0.027

4.5. CROSS LINE ANALYSES

A crossline was run in a direction perpendicular to main scheme lines, providing a good representation for analysis of consistency.

Crossline analysis was performed using the Fledermaus CrossCheck tool. Crossline point data were compared to a 0.5m gridded surface of the main scheme survey lines and statistics generated. A histogram of the point comparison was reviewed in CrossCheck to ensure there was a normal distribution of data.

A summary of the CrossCheck results is provided in Table 13. Areas of increased surface roughness and steep slopes were removed from the analysis, where acceptable differences in horizontal position can affect vertical comparisons. Therefore, cross checks were run using only areas with slopes of less than 10 degrees.

Table 13: Cross Line Point to Surface Results

No. of Points Compared	2 059 899
Mean Difference (MD) in m	0.00
Standard Deviation (St. Dev)	0.05
Mean + 2* Std. Dev	0.09

In addition, 0.5m surfaces were created for the crosslines, and surface differences generated between the crossline and main scheme surfaces. Statistics for the difference surfaces were generated. Results matched those from the CrossCheck analysis, as shown below.

Table 14: Cross Line Surface to Surface Results

Mean Difference (MD) in m	0.00
Standard Deviation (St. Dev)	0.08

4.6. VERTICAL ACCURACY CHECKS

4.6.1. COMPARISON TO TOPO CONTROL

Topo control data were acquired in Niue using RTK GNSS techniques.

Terrascan was then used to compare the lidar data to known ground control points. For each known location a small TIN was created from the surrounding lidar points and the elevation difference from the TIN plane to the point computed. Data shows good agreement with the topo control (Table 15).

Table 15: Comparison to Topo Control Results

	Survey Data
Average dZ (m)	0.001
Std. Deviation	0.023
RMSE (m)	0.023

5. PRODUCTS

Deliverables required for the project are listed in Table 16, and described below.

Table 16: Project Deliverables

ID	Item	Deliverable	Data Drive Folder
1	Classified Point Cloud	LAS 1.4 Flight Line Files on the Ellipsoid	01_LAS\Ellipsoid
		LAS 1.4 Flight Line Files on Sounding Datum	01_LAS\SoundingDatum
2	Bare Earth DEM	50cm Geotiff of DEM on the Ellipsoid	02_DEM
		50cm Geotiff of DEM on Sounding Datum	02_DEM
3	Digital Surface Model (DSM)	50cm Geotiff of DSM on the Ellipsoid	03_DSM
		50cm Geotiff of DSM on the Sounding Datum	03_DSM
4	Aerial Image Mosaics	10cm 4-Band (RGBN) tiled Geotiffs	04_Imagery\GeoTIFF
		10cm 3-Band (RGB) tiled ECWs	04_Imagery\ECW

All data and deliverables are provided in the following datum and projection:

- Horizontal: WGS84, UTM Zone 02S, meters
- Vertical: Sounding Datum, meters

A copy of the LAS files, DEM, and DSM was also delivered with vertical on the ellipsoid.

All times in the LAS files and throughout the survey are in UTC.

5.1. CLASSIFIED POINT CLOUD (LAS FILES)

The classified point cloud LAS 1.4 files are delivered using standard ASPRS Classification Levels, with the LAS Topo-Bathy Domain Profile (July 17, 2013) classes. Final LAS classes included are provided in Table 17.

All LAS files include intensity values and are colorized from the image mosaics. Therefore, all data are in Point Record Format 8.

Table 17: Delivered LAS Classes

CLASS	Description
1	Unclassified
2	Ground
7	Low Point (noise)
9	Water
18	High Noise
40	Bathymetric point (e.g., seafloor or riverbed; also known as submerged topography)
41	Water surface (sea/river/lake surface from bathymetric or topographic-bathymetric lidar; distinct from Point Class 9, which is used in topographic-only lidar and only designates "water," not "water surface")
42	Derived water surface (synthetic water surface location used in computing refraction at water surface)
45	No-bottom-found-at (bathymetric lidar point for which no detectable bottom return was received)
47	Bathy Noise
48	Bathy Land

APPENDIX A : FLIGHT LOGS

PROJECT NAME:	80522 - Niue - TopoBathy Lidar	BASE AIRPORT:	Niue (NIUE)
LOCATION / AREA:	Niue / BL01	DATE:	7 January 2020
AIRCRAFT:	Reims 406 - ZK-XLF	PILOT:	Cole B.
SYSTEM:	Chiroptera 4X	OPERATOR:	Dushan A.
MISSION ID:	80522_Niue	CLOUDS:	Clouds @ 1800
BASE STATION:	RM1	WIND:	10-15 kts @ 110
ENGINE START:	18:16	ENGINE OFF:	19:57
GNSS START:		ENGINE TIME:	01:41
TAKEOFF:	18:42	GNSS START:	
		TOUCHDOWN:	19:53
		AIR TIME	01:11

FL #	LINE #	START TIME	END TIME	TOPO PRF PWR		CHII PWR	REMARKS
		18:54:55					DS: Topo_20200107_185510
000_FL3	0103	18:55:10	18:58:06	500	13	295	
001_FL2	0102	19:00:16	19:03:02	500	13	295	
002_FL1	0101	19:05:42	19:08:15	500	13	295	
003_FL4	0104	19:10:56	19:14:08	500	13	295	
004_FL5	0105	19:16:07	19:19:30	500	13	295	
005_FL6	0106	19:21:25	19:24:48	500	13	295	
006_FL7	0107	19:28:08	19:33:04	500	13	295	
007_FL8	0108	19:35:10	19:40:09	500	13	295	
008_FL9	0109	19:42:18	19:46:07	500	13	295	
		19:50:00					Abort mission: Clouds

PROJECT NAME:	80522 - Niue - TopoBathy Lidar	BASE AIRPORT:	Niue (NIUE)
LOCATION / AREA:	Niue / BL01	DATE:	8 January 2020
AIRCRAFT:	Reims 406 - ZK-XLF	PILOT:	Cole B.
SYSTEM:	Chiroptera 4X	OPERATOR:	Dushan A.
MISSION ID:	80522_Niue	CLOUDS:	Clouds @ 1800
BASE STATION:	RM1	WIND:	15-20 kts @ 120
ENGINE START:	17:22	ENGINE OFF:	22:18
ENGINE TIME:			04:56
GNSS START:		GNSS START:	
TAKEOFF:	17:34	TOUCHDOWN:	22:14
AIR TIME			04:40

FL #	LINE #	START TIME	END TIME	TOPO PRF PWR		CHII PWR	REMARKS
		17:40:01					DS: Topo_20200108_174016
000_FL30	0130	17:40:16	17:46:41	500	13	295	
001_FL31	0131	17:49:09	17:55:54	500	13	295	
002_FL32	0132	17:57:46	18:04:30	500	13	295	
003_FL33	0133	18:06:32	18:13:10	500	13	295	
004_FL34	0134	18:15:37	18:22:29	500	13	295	
005_FL35	0135	18:24:32	18:31:01	500	13	295	
006_FL36	0136	18:32:57	18:39:06	500	13	295	
007_FL37	0137	18:40:55	18:46:59	500	13	295	
008_FL38	0138	18:49:06	18:55:09	500	13	295	
009_FL39	0139	18:57:20	19:03:21	500	13	295	
010_FL40	0140	19:05:21	19:11:11	500	13	295	
011_FL41	0141	19:12:51	19:18:35	500	13	295	
012_FL42	0142	19:20:20	19:26:10	500	13	295	
013_FL43	0143	19:27:43	19:33:22	500	13	295	
014_FL44	0144	19:34:52	19:40:34	500	13	295	
015_FL45	0145	19:42:15	19:48:08	500	13	295	
016_FL46	0146	19:49:43	19:55:18	500	13	295	
017_FL47	0147	19:57:08	20:02:40	500	13	295	
018_FL48	0148	20:04:31	20:09:46	500	13	295	
019_FL49	0149	20:11:35	20:16:59	500	13	295	
020_FL50	0150	20:18:55	20:24:06	500	13	295	
021_FL51	0151	20:26:13	20:31:27	500	13	295	
022_FL52	0152	20:33:14	20:38:26	500	13	295	
023_FL53	0153	20:40:29	20:45:46	500	13	295	
024_FL54	0154	20:47:41	20:52:43	500	13	295	
025_FL55	0155	20:55:05	21:00:01	500	13	295	
026_FL56	0156	21:01:57	21:06:44	500	13	295	
027_FL57	0157	21:08:40	21:13:28	500	13	295	
028_FL58	0158	21:15:34	21:20:03	500	13	295	

PROJECT NAME: 80522 - Niue - TopoBathy Lidar
LOCATION / AREA: Niue / BL01
AIRCRAFT: Reims 406 - ZK-XLF
SYSTEM: Chiroptera 4X

BASE AIRPORT: Niue (NIUE)
DATE: 8 January 2020
PILOT: Cole B.
OPERATOR: Dushan A.

MISSION ID: 80522_Niue
BASE STATION: RM1

CLOUDS: Clouds @ 1800
WIND: 15-20 kts @ 120

ENGINE START: 17:22 **ENGINE OFF:** 22:18 **ENGINE TIME:** 04:56
GNSS START: **GNSS START:**
TAKEOFF: 17:34 **TOUCHDOWN:** 22:14 **AIR TIME** 04:40

FL #	LINE #	START TIME	END TIME	TOPO PRF PWR		CHII PWR	REMARKS
029_FL59	0159	21:21:46	21:26:14	500	13	295	
030_FL60	0160	21:27:48	21:32:08	500	13	295	
031_FL61	0161	21:33:44	21:38:02	500	13	295	
032_FL62	0162	21:39:45	21:43:51	500	13	295	
033_FL63	0163	21:45:33	21:48:51	500	13	295	
034_FL64	0164	21:50:44	21:53:37	500	13	295	
035_FL65	0165	21:56:02	21:58:22	500	13	295	
036_FL66	0166	22:00:19	22:02:09	500	13	295	
037_FL67	0195	22:04:43	22:09:52	500	13	295	

PROJECT NAME:	80522 - Niue - TopoBathy Lidar	BASE AIRPORT:	Niue (NIUE)
LOCATION / AREA:	Niue / BL01	DATE:	8 January 2020
AIRCRAFT:	Reims 406 - ZK-XLF	PILOT:	Cole B.
SYSTEM:	Chiroptera 4X	OPERATOR:	Dushan A.

MISSION ID:	80522_Niue	CLOUDS:	Clouds @ 1800
BASE STATION:	RM1	WIND:	15-20 kts @ 100

ENGINE START:	22:45	ENGINE OFF:	1:54	ENGINE TIME:	03:09
GNSS START:		GNSS START:			
TAKEOFF:	22:55	TOUCHDOWN:	1:50	AIR TIME	02:55

FL #	LINE #	START TIME	END TIME	TOPO PRF PWR		CHII PWR	REMARKS
		22:59:09					DS: Topo_20200108_225924
000_FL29	0129	22:59:24	23:05:38	500	13	295	
001_FL28	0128	23:07:40	23:13:43	500	13	295	
002_FL27	0127	23:15:37	23:21:38	500	13	295	
003_FL26	0126	23:23:48	23:29:46	500	13	295	
004_FL25	0125	23:31:44	23:37:39	500	13	295	
005_FL24	0124	23:39:41	23:45:36	500	13	295	
006_FL23	0123	23:47:50	23:53:36	500	13	295	
007_FL22	0122	23:56:01	00:01:47	500	13	295	
008_FL21	0121	00:04:13	00:09:58	500	13	295	
009_FL20	0120	00:12:12	00:17:52	500	13	295	
010_FL19	0119	00:19:51	00:25:35	500	13	295	
011_FL18	0118	00:27:43	00:33:16	500	13	295	
012_FL17	0117	00:35:17	00:41:01	500	13	295	
013_FL16	0116	00:43:14	00:48:45	500	13	295	
014_FL15	0115	00:50:43	00:56:14	500	13	295	
015_FL14	0114	00:58:28	01:03:58	500	13	295	
016_FL13	0113	01:06:14	01:11:50	500	13	295	
017_FL12	0112	01:14:20	01:19:42	500	13	295	
018_FL11	0111	01:21:53	01:27:14	500	13	295	
019_FL10	0110	01:29:36	01:34:59	500	13	295	
020_FL9	0109	01:37:10	01:42:23	500	13	295	
021_FL3	0103	01:44:27	01:47:32	500	13	295	

PROJECT NAME:	80522 - Niue - TopoBathy Lidar	BASE AIRPORT:	Niue (NIUE)
LOCATION / AREA:	Niue /	DATE:	9 January 2020
AIRCRAFT:	Reims 406 - ZK-XLF	PILOT:	Cole B.
SYSTEM:	Chiroptera 4X	OPERATOR:	Dushan A.
MISSION ID:	Cal-Niue	CLOUDS:	Clouds @ 1800
BASE STATION:	RM1	WIND:	15-20 kts @ 85
ENGINE START:	21:07	ENGINE OFF:	22:28
ENGINE TIME:			01:21
GNSS START:		GNSS START:	
TAKEOFF:	21:23	TOUCHDOWN:	22:24
AIR TIME			01:01

FL #	LINE #	START TIME	END TIME	TOPO PRF PWR	CHII PWR	REMARKS
		21:26:00				Deep W:-150 / G:-70
		21:28:14				DS: 500m_20200109_212829
000_FL1	0501	21:28:29	21:30:04	500	13	40
001_FL2	0502	21:32:28	21:34:05	500	13	40
002_FL3		21:36:31	21:36:33			BAD: Safety Shutoff
003_FL3	0503	21:40:39	21:42:15	500	13	40
004_FL4	0504	21:44:47	21:46:22	500	13	40
005_FL5	0505	21:48:46	21:50:19	500	13	40
006_FL6		21:55:38	21:55:39			BAD: Safety Shutoff
007_FL6	0506	21:55:54	21:57:15	500	13	40
		21:58:00				Deep W:-140 / G:-60
		21:59:46				DS: 400m_20200109_220001
000_FL1	0401	22:00:01	22:01:34	300	10	30
001_FL2	0402	22:03:27	22:05:00	300	10	30
002_FL3	0403	22:07:10	22:08:33	300	10	30
003_FL4	0404	22:10:46	22:12:15	300	10	30
004_FL5	0405	22:14:16	22:15:46	300	10	30
005_FL6	0406	22:19:18	22:20:45	300	10	30

APPENDIX B : PROCESSING LOGS

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

MISSION ID: 80522_Niue **LINES:** 67
SURVEY BLOCKS: BL01 **LINE KM:** 1,192

Flight	Engine Time	Air Time	Flown km %		Reflown km %		Comments
2020-01-07A	1:41:00	1:11:00	110.7	9.3%			BL01
2020-01-08A	4:56:00	4:40:00	680.5	57.1%			BL01
2020-01-08B	3:09:00	2:55:00	400.9	33.6%	27.8	2.3%	BL01
2020-01-09A	1:21:00	1:01:00					Niue Calibration
TOTAL	11:07:00	9:47:00	1192	100.0%	28	2.3%	

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

Axis	Ref to IMU In the system	Ref to GNSS In the system	IMU to Gimble Inertial Explorer	IMU to GNSS Inertial Explorer	Gimble to GNSS Inertial Explorer	Differences from Previous	Differences from Start
Start Values						Flight Pro	Inertial Explorer
X	-0.003	0.006	0.005	-0.060	-0.065	+ tve forward	+ tve stbd
Y	-0.006	0.000	0.003	0.600	0.597	+ tve stbd	+ tve forward
Z	-0.296	-1.305	-0.296	0.930	1.226	+ tve down	+ tve up
Lever Arm Estimate in Inertial Explorer - Test 1							
X		0.604	0.005	-0.056	-0.061	0.004	0.004
Y		-0.061	0.003	0.607	0.604	0.007	0.007
Z		-1.220	-0.296	0.924	1.220	-0.006	-0.006
Lever Arm Estimate in Inertial Explorer - Test 2							
X		0.608	0.005	-0.054	-0.059	0.002	0.006
Y		-0.059	0.003	0.611	0.608	0.004	0.011
Z		-1.220	-0.296	0.924	1.220	0.000	-0.006
Lever Arm Estimate in Inertial Explorer - Test 3							
X		0.610	0.005	-0.053	-0.058	0.001	0.007
Y		-0.058	0.003	0.613	0.610	0.002	0.013
Z		-1.220	-0.296	0.924	1.220	0.000	-0.006
Lever Arm Estimate in Inertial Explorer - Test 4							
X		0.611	0.005	-0.052	-0.057	0.001	0.008
Y		-0.057	0.003	0.614	0.611	0.001	0.014
Z		-1.220	-0.296	0.924	1.220	0.000	-0.006
Lever Arm Estimate in Inertial Explorer - Test 5							
X	-0.003	0.611	0.005	-0.052	-0.057	0.000	0.008
Y	-0.006	-0.057	0.003	0.614	0.611	0.000	0.014
Z	-0.296	-1.220	-0.296	0.924	1.220	0.000	-0.006

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

File	Antenna Height (m)	Antenna Type	Ephemeris	Date	Time		N Latitude			W Longitude			Height (m)	RMS (m)	Comments
					Start	Stop	Deg	Min	Sec	Deg	Min	Sec			
Station: XXXX															
RM1_74161780	1.550	R10	RTX	2018-06-27	19:52	0:42	19	4	35.10213	169	55	37.27603	88.092	0.013	Trimble-RTX, ITRF14 (2018.47)
RM1_74161791	1.550	R10	RTX	2018-06-28	0:47	3:20	19	4	35.10214	169	55	37.27604	88.100	0.017	Trimble-RTX, ITRF14 (2018.47)
RM1_74161792	1.550	R10	RTX	2018-08-28	19:17	0:29	19	4	35.10227	169	55	37.27573	88.089	0.011	Trimble-RTX, ITRF14 (2018.49)
RM1_74161840	1.550	R10	RTX	2018-07-03	19:05	23:34	19	4	35.10310	169	55	37.27622	88.128	0.011	Trimble-RTX, ITRF14 (2018.49)
		Average:					19	04	35.10241	169	55	37.27601	88.102		
		Standard Deviation:								0.00040			0.00018		

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

PID	Project Name	Latitude			Longitude			Height (m)	XY Datum	Z Datum	Source	Comments
		Deg	Min	Sec	Deg	Min	Sec					
RM1		19	4	35.10241	169	55	37.27601	88.102	ITRF2014	ITRF2014	RTX	

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project /Raw	Run Project Wizard	Base Station			Check Base Coordinate & Datum	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2020-01-07A_CH4X	DA	DA	DA	DA	RM1	R10	1.800	DA	DA			DA	DA	0.008	0.022	0.035	Final	
2020-01-08A_CH4X	DA	DA	DA	DA	RM1	R10	1.800	DA	DA			DA	DA	0.009	0.020	0.031	Final	
2020-01-08B_CH4X	DA	DA	DA	DA	RM1	R10	1.800	DA	DA			DA	DA	0.013	0.007	0.049	Final	
2020-01-09A_CH4X	DA	DA	DA	DA	RM1	R10	1.800	DA	DA			DA	DA	0.007	0.004	0.013	Final	
2020-01-09A_HE4X	DA	DA	DA	DA	RM1	R10	1.800	DA	DA			DA	DA	0.006	0.004	0.015	Final	

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

Mission	Copied to Disk	Nav Session	Nav Type	Calibration File	Cal Type	Processing Parameters	Check Processing Parameters	Processing Session	Number of FL	Process Topo	Process Shallow	Process Deep	Run QC Stats in FME	Comments
Cal-Niue_400m_20200109_220001	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r0	Prelim	ProcessingCal_20191002_400m	MW	r0/400m_20200210_080457	6	MW	MW	MW	MW	Topo, Shallow, Deep
Cal-Niue_500m_20200109_212829	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r0	Prelim	ProcessingCal_20191002_500m	MW	r0/500m_20200210_080501	6	MW	MW	MW	MW	Topo, Shallow, Deep
								Calibration_2020-02-10_11.11.16						Topo - Update Angles (r1)
								Calibration_2020-02-10_11.12.04						Shallow - Update Angles (r1)
								Calibration_2020-02-10_11.12.26						Deep - Update Angles (r1)
														Topo, Shallow and Deep Slant Range are updated.
														Updated Cal File for Topo, Shallow, Deep Angles and Slant Range (r1)
Cal-Niue_400m_20200109_220001	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_400m	MW	r1/400m_20200210_173435	6	MW	MW	MW	MW	Topo, Shallow, Deep
Cal-Niue_500m_20200109_212829	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_500m	MW	r1/500m_20200210_173427	6	MW	MW	MW	MW	Topo, Shallow, Deep
								Calibration_2020-02-10_20.37.20						Topo - Update Angles (r2)
								Calibration_2020-02-10_20.38.00						Shallow - Update Angles (r2)
								Calibration_2020-02-10_20.40.34						Deep - Update Angles (r2)
														Topo and Deep Slant Range are locked. Shallow Slant Range is updated
														Updated Cal File for Topo, Shallow, Deep Angles and Slant Range (r2)
Cal-Niue_400m_20200109_220001	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_400m	MW	r2/400m_20200210_212423	6	MW	MW	MW	MW	Topo, Shallow, Deep
Cal-Niue_500m_20200109_212829	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_500m	MW	r2/500m_20200210_212420	6	MW	MW	MW	MW	Topo, Shallow, Deep
								Calibration_2020-02-10_23.46.37						Topo - Update Angles (r3)
								Calibration_2020-02-10_23.46.53						Shallow - Update Angles (r3)
								Calibration_2020-02-10_23.47.03						Deep - Update Angles (r3)
														Topo Shallow and Deep Slant Range are locked.
														Updated Cal File for Topo, Shallow, Deep Angles (r3)
Cal-Niue_400m_20200109_220001	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_400m	MW	r3/400m_20200211_061249	6	MW	MW	MW	MW	Topo, Shallow, Deep
Cal-Niue_500m_20200109_212829	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_500m	MW	r3/500m_20200211_061247	6	MW	MW	MW	MW	Topo, Shallow, Deep
								20200211_075640_MirrorCalibration						Topo - Mirror Cal (r4)
								20200211_075641_MirrorCalibration						Shallow - Mirror Cal (r4)
								20200211_075642_MirrorCalibration						Deep - Mirror Cal (r4)
														Topo Shallow and Deep Slant Range are locked.
														Updated Cal File for Topo, Shallow, Deep Mirror Cal (r4)
Cal-Niue_400m_20200109_220001	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_400m	MW	r4/400m_20200211_082112	6	MW	MW	MW	MW	Topo, Shallow, Deep
Cal-Niue_500m_20200109_212829	MW	2020-01-09A	Final	CAL_TSD_Survey_20191002_r1	Interim	ProcessingCal_20191002_500m	MW	r4/500m_20200211_082115	6	MW	MW	MW	MW	Topo, Shallow, Deep
r4 Renamed to CAL_TSD_Survey_2020107_r0 for Project Processing														

PROJECT NAME: 080522 - Niue - Topo Lidar
LOCATION: Niue
AIRCRAFT: Reims F406 - ZK-XLF
SYSTEM: Chiroptera 4X

Mission	Copied to Disk	Nav Session	Nav Type	Calibration File	Cal Type	Processing Parameters	Check Processing Parameters	Processing Session	Number of FL	Process Topo	Process Shallow	Process Deep	Verify Derived Water Surface (Coverage and Height)	Review Submerged Data for any settings issues	Review Shallow to Deep, Shallow to Topo	Colorize in LSS	Review FL SHP	Export Trajectories	Merge in FME	Comments
Niue_Topo_20200107_185510	DA	2020-01-07A	Final	CAL_TSD_Survey_20200107_r0	Final	ProcSurvey_20200324_500m_Topo_r0	BH	20200107_185510	9	9	9	--	AB	AB	AB	--	AB	X	AB	
Niue_Topo_20200108_174016	DA	2020-01-08A	Final	CAL_TSD_Survey_20200107_r0	Final	ProcSurvey_20200324_500m_Topo_r0	BH	20200108_174016	38	38	38	--	AB	AB	AB	--	AB	X	AB	
Niue_Topo_20200108_225924	DA	2020-01-08B	Final	CAL_TSD_Survey_20200107_r0	Final	ProcSurvey_20200324_500m_Topo_r0	BH	20200108_225924	22	22	22	--	AB	AB	AB	--	AB	X	AB	

PROJECT NAME: 80522 - Niue - TopoBathy Lidar
LOCATION: Niue
AIRCRAFT: Reims-Cessna 406 - ZK-XLF
SYSTEM: Chiropter 4X

RCD30 Dataset	HXMAP					IPAS CO+					Estimate Misalignment												Comments
	Ingest RCD30 Data	Assign Radio Corrections	Export Images	Export IPAS Files	Bits	IPAS Solution	Camera File	PPO X (mm)	PPO Y (mm)	Camera File Status	APM	Run AT	Sigma0	PPA X (mm)	PPA Y (mm)	Misalign X (arcmin)	Misalign Y (arcmin)	Misalign Z (arcmin)	Misalign X RMS (arcmin)	Misalign Y RMS (arcmin)	Misalign Z RMS (arcmin)	Accept / Reject	
2020-01-09A	TH	TH	TH	TH	8	2020-01-09A	IPAS_RCD30_82541	0.0000	0.0000	Initial	5x5	TH	22.6	0.0000	0.0000	-0.330	0.240	-0.510	0.310	0.290	0.690	accept	500m AGL
							IPAS_RCD30_82541_r1	0.0000	0.0000	Interim	5x5	TH	18.4	-0.0200	0.0000	-0.016	0.050	-0.160	0.240	0.240	0.600	Reject	
Exported to: RCD30_Geometry_CameraHead-82541-D-798528_LensSystem-50149--785422_DateTime-20200319-161522																							
2020-01-09A	TH	TH	TH	TH	8	2020-01-09A	IPAS_RCD30_82541	0.0000	0.0000	Final	5x5	TH	18.2										500m AGL, RMSE = 0.048682

PROJECT NAME: 80522 - Niue - TopoBathy Lidar
LOCATION: Niue
AIRCRAFT: Reims-Cessna 406 - ZK-XLF
SYSTEM: Chiropter 4X

RCD30 Dataset	HXMAP							FOR LSS		OrthoVista			FME	Comments
	Ingest	QC Radio Profile	Assign Radio Profile	Correction Type	DTM Ready	Create Block	Export Ortho	Export IPAS	Export Aerial Images	Mosaic by Flight Session (10cm)	Mosaic Tiles (10cm)	Stretched Tiles	ECW Created	
20200107-A1	TH	1.5	TH	CAL	TH	TH	TH	--	--	--				Lines 01 to 09 (Line 09 not used, short due to clouds)
20200108-A1	TH	1.6	TH	CAL	TH	TH	TH	--	--	--				Lines 30 to 66
20200108-B1	TH	1.3	TH	CAL	TH	TH	TH	--	--	--				Lines 03, 09 to 29 (Line 03 not used, reflight)
20200109-A1														cal